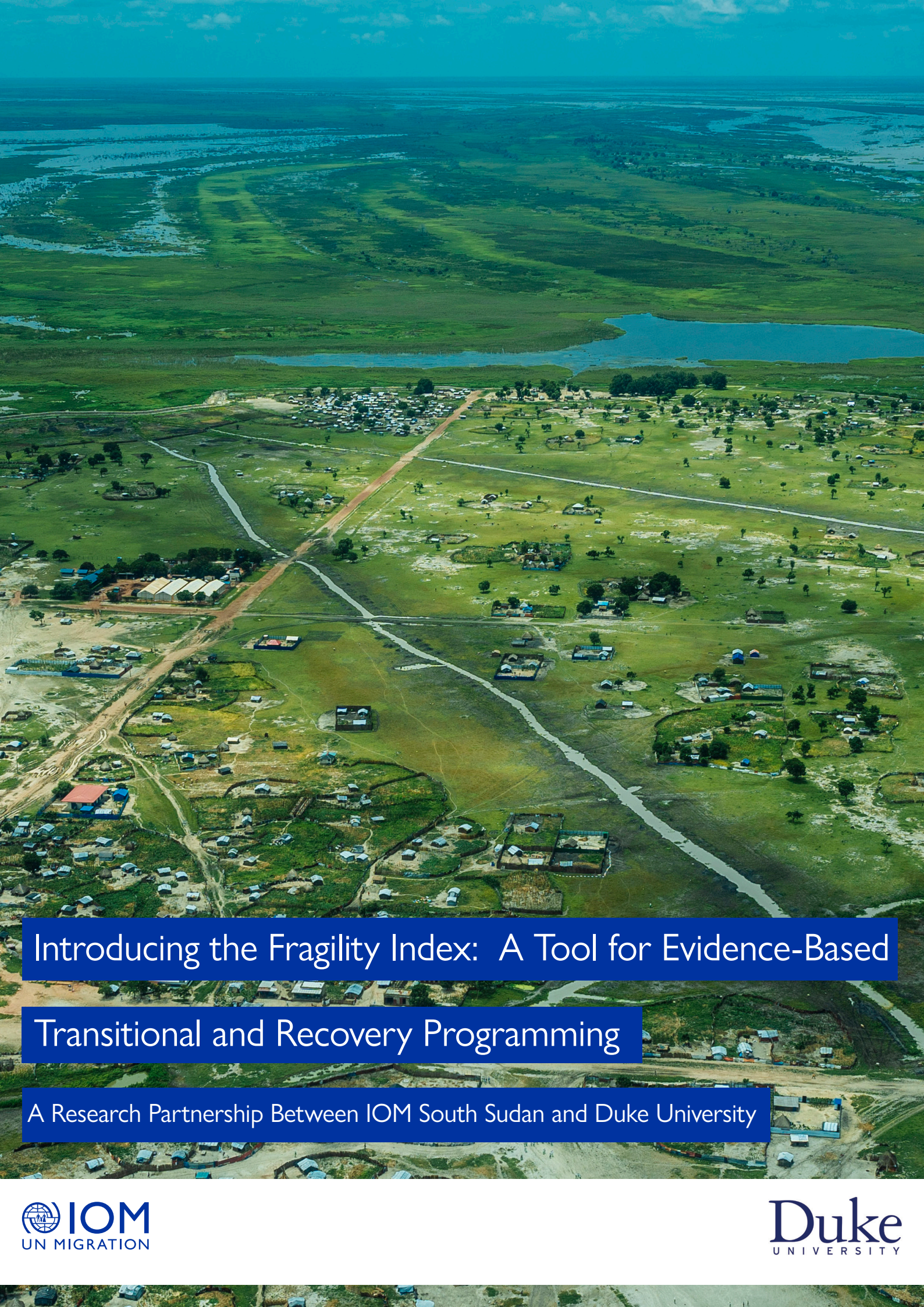




Introducing the Fragility Index: A Tool for Evidence-Based Transitional and Recovery Programming

A Research Partnership Between IOM South Sudan and Duke University

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A Research Partnership Between IOM South Sudan and Duke University

Preview Prepared for October 30, 2024

Gabriella Levy, Ph.D. and Mara Revkin, Ph.D.¹

1. This summary was prepared by Dr. Gabriella Levy, Assistant Professor of Political Science at the University of Washington and Dr. Mara Revkin, Associate Professor of Law and Political Science at Duke University. Authors are listed alphabetically and contributed equally.

TABLE OF CONTENTS

Overview.....5

Fragility Index Methodology and Objectives5

Defining and Measuring Multidimensional Fragility..... 7

Complementarities with Existing Assessment Tools and Added Value 8

How to Interpret the Index and Preliminary Findings 8

Implications for Programming and Targeting 10

Recommendations and Next Steps for the Fragility Index 12



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OVERVIEW

The International Organization for Migration (IOM) in South Sudan has partnered with an interdisciplinary research team at Duke University to develop a novel Fragility Index, designed to support humanitarian and development actors in moving beyond emergency response programming towards transitional and recovery programming. The Fragility Index, a new addition to IOM's global Data for Solutions (D4S) initiative,² is a valuable tool for evidence-based programming that can be used at design, targeting, and impact evaluation stages. This summary previews the findings of a longer forthcoming report based on a proof-of-concept pilot survey, which gathered data from a random sample of 3,685 respondents across nine counties in South Sudan conducted in April-May 2024,³ building on a smaller pilot study conducted in June-July 2023.⁴

FRAGILITY INDEX METHODOLOGY AND OBJECTIVES

The concept of “fragility” is defined by the Organisation for Economic Co-operation and Development (OECD) as “the combination of exposure to risk and insufficient coping capacities to manage, absorb and mitigate those risks.”⁵ Fragility has received growing attention in the context of the UN Secretary-General's 2017 calls for prioritizing conflict prevention and connecting the pillars of development, peace and security, and human rights⁶ as well as amidst the World Bank's revised 2020-2025 strategy on fragility, conflict, and violence. This strategy emphasizes a “pivot” to prevention.⁷

The Fragility Index operationalizes six dimensions of fragility - societal, economic, political/legal, security, environmental, and human - with household-level survey data, providing nuanced snapshots of local vulnerabilities and capacities at different geographical scales, including the household, the village, and the county level. In doing so, it generates high-quality data for solutions and informs strategic, evidence-based decisions that support equitable resource allocation and foster resilience in South Sudanese communities. Each of the six dimensions

is measured by five simple questions. The Index can support evidence-based programming in line with the Humanitarian Development Peace Nexus (HDPN) in the following ways:

- It provides a comprehensive **assessment of vulnerabilities and capacities among the general population that can be disaggregated** by household displacement status, rurality, and more;
- produces fine-grained data on variation in different dimensions of fragility (economic, environmental, political/legal, security, societal) to inform the **design** of context-specific, coordinated, multi-sectoral programs covering the pillars of recovery, stabilization, and reintegration, as well as the **targeting** of this programming based on objective, evidence-based criteria;
- can measure **changes in different dimensions of fragility over time**;

2. International Organization for Migration Displacement Tracking Matrix (IOM DTM), “[Data for Solutions](#),” (Geneva, IOM).

3. See Appendix D for a full list of sampled bomas.

4. Olga Aymerich, Mara Revkin and Samuel Akau, “[Understanding Multidimensional Fragility in South Sudan](#),” (Juba, IOM DTM, 2023).

5. The Organisation for Economic Co-operation and Development (OECD), *States of Fragility 2022*, (Paris, OECD, 2022).

6. United Nations Security Council, “[Secretary-General, in First Address to Security Council Since Taking Office, Sets Restoring Trust, Preventing Crises as United Nations Priorities](#),” (New York, Security Council Meetings Coverage, 2017). See also United Nations News, “[Security Council has critical role in addressing fragility and conflict: UN chief](#),” (New York, United Nations News, 2021).

7. World Bank, *World Bank Group Strategy for Fragility, Conflict, and Violence 2020–2025*, (Washington, D.C., World Bank, 2020).

- can be adapted for data collection in contexts beyond South Sudan;⁸
- and can be linked to a range of different outcomes and objectives across the pillars of peace and security, human rights, and development, potentially providing the foundations for **baseline and endline impact evaluations** of specific programmatic interventions that seek to reduce one or more dimensions of fragility.

The Fragility Index was developed in the context of calls for more evidence on the effectiveness of different targeting approaches and how they are perceived by beneficiaries. In recent years, several UN authorities and agencies have raised concerns about potential unintended harmful consequences of status-based targeting approaches, in which aid or programming is provided to internally displaced persons (IDPs), women, children, or other sub-groups based on their membership in that category rather than their individual needs. One concern is that status-based aid can exacerbate intergroup tensions and competition. For example, the 2019 report of the UN Secretary General on South Sudan indicated that providing assistance to people in Protection of Civilian (PoC) sites based solely on their residential status “raises questions of equity.”⁹ Previous research in other contexts including Zimbabwe, Ethiopia, and Somalia has similarly found that status-based assistance “can exacerbate tensions through perceived favorable treatment of certain groups.”¹⁰ A second concern is that status-based targeting may not lead to the most efficient or fair use of resources in contexts of extreme need and limited funding. In light of these concerns, WFP and UNHCR in South Sudan recently announced plans to shift from status-based food and non-food items (NFI)

assistance to targeted and prioritized assistance based on refugee households’ vulnerability levels to ensure an efficient use of the limited resources available.¹¹

In this context of ongoing debates about how to improve aid targeting amidst limited resources, the Fragility Index can directly inform all three leading approaches to assessing needs and allocating aid:

1. **Geographic/Area-Based:** By allowing for comparisons of relative levels of a particular dimension of fragility across different geographic areas, the Fragility Index can strengthen area-based targeting. For example, areas where human fragility scores are highest should have the greatest need for life-saving humanitarian aid including WASH, food assistance, and shelter. Conversely, areas with the lowest scores on the economic fragility dimension may be most prepared to benefit from development programs such as loans for small and medium-sized enterprises.
2. **Status-Based:** The Fragility Index provides detailed data on variation in the vulnerabilities of individuals within a particular status-based group that can be used to improve the precision of targeting within that group. For example, it can indicate which IDPs are most in need of shelter. It can also provide concrete evidence in support of or against the need for status-based targeting for certain groups of potential beneficiaries.
3. **Household-Based:** The Fragility Index provides a comprehensive but simple measure of household needs and capacities that can be used to allocate individualized support such as cash assistance.

8. While each question corresponds to a concept which is universal, not all question wording is applicable to all contexts. See notes regarding where question wording may need to be adjusted in Appendix C.

9. United Nations Secretary General, “S/2019/741 Future planning for the protection of civilians sites in South Sudan,” (New York, United Nations Security Council, 2019).

10. Sam Grundy and Sarah Zingg, “MRS No. 66 - Community Stabilization – An approach for facilitating progress towards durable solutions and operationalizing the Humanitarian–Development–Peace Nexus: Lessons from Zimbabwe, Ethiopia and Somalia,” (Geneva, IOM, 2020).

11. United Nations High Commissioner for Refugees (UNHCR) and World Food Programme (WFP), “Prioritisation of assistance in South Sudan: Community consultation report,” (Geneva, UNHCR, 2023).

DEFINING AND MEASURING MULTIDIMENSIONAL FRAGILITY

This study adapts the OECD multidimensional fragility framework for use at the community level and tailors it for the context of South Sudan.¹² These different dimensions are defined as follows:

1. **Economic Fragility** relates to the vulnerabilities and challenges within the economic systems of a country or region. In the Fragility Index, this component includes measures of education, financial satisfaction, the ability to pay for expected and unexpected large costs, and dependency on aid.
2. **Environmental Fragility** concerns the challenges and risks associated with the environment and natural resources. In the Fragility Index, this component includes climate change's impact on income, climate change's impact on food consumption, climate change's impact on mobility, climate change's impact on health, and the ability to act on early warning information to ensure household safety.
3. **Human Fragility** refers to people's "well-being and ability to live healthy, long and prosperous lives."¹³ The Fragility Index focuses on factors relevant for health: food access, water access, medical treatment access, shelter, and sexual violence.
4. **Political and Legal Fragility**¹⁴ refers to the instability, weaknesses, and challenges in political systems and governance structures. In the Fragility Index, this component encompasses trust in state government, trust in national government, trust in informal authorities, participation in democracy without fear, and freedom of speech.
5. **Security Fragility** refers to the presence of threats, conflicts, and violence within a society. In the Fragility Index, this component includes the degree to which individuals are affected by crime, war, land conflict, resource conflict, and local armed militias.
6. **Societal Fragility** focuses on societal divisions, inequalities, and social cohesion. In the Fragility Index, it includes experiences of acceptance, experiences of discrimination, social cohesion, network embeddedness, and migrant-local cohesion.

12. The Organisation for Economic Co-operation and Development (OECD), *States of Fragility 2022*, (Paris, OECD, 2022).

13. The Organisation for Economic Co-operation and Development (OECD), *"Adding the Human Dimension to the OECD Fragility Framework"* (Paris, OECD, 2022). OECD includes a range of health indicators as well as human capital indicators, such as school enrollment.

14. OECD's fragility framework does not include a standalone dimension for "legal" fragility. This project has modified the category of "political fragility" to "political and legal fragility" to capture important aspects of the legal context including rule of law and preferences for dispute resolution among alternative providers of justice including state courts and customary tribal or religious authorities.

Complementarities with Existing Assessment Tools and Added Value:

The Fragility Index is unique in that it utilizes household-level survey data to provide a representative mapping of vulnerabilities and needs at the population level within a particular area. However, the Fragility Index complements existing assessment tools within the D4S initiative. For example, the Migration Integration Toolkit (MIT) facilitates status-based targeting of aid for migrants¹⁵ Although MIT provides valuable individual-level data on several dimensions of integration (social, political, navigational, economic, psychological and linguistic), it is limited to a particular sub-group: migrants. As a result, unlike the Fragility Index, it provides only limited insight into the larger communities in which migrants live. Two other tools, the Stability Index (SI) and the Solutions Mobility Index (SMI),¹⁶ use perception data to operationalize varying components of stability, depending on the country – these components are (1) safety and security, (2) livelihoods and basic services, (3) social cohesion, and (4) disaster adaptation. Unlike the Fragility Index, both tools are based on interviews with non-random samples of “key informants”¹⁷ who represent important social groups or other constituencies, including “community leaders, aid workers, and other community representatives.”¹⁸ Although key informant interviews provide valuable qualitative insights into local dynamics, they can be susceptible to biases that reflect the demographics and positionality of the key informants, who are often disproportionately older male elites.¹⁹ For example, the Solutions and Mobility Index Round 1 in Ghana conducted 395 key informant interviews, of which 91 per cent were with men.²⁰ The Fragility Index relies on representative random samples in which all eligible adults living in a particular area have an equal

probability of being selected for participation. These two sources of information – qualitative data from key informant interviews and quantitative data from randomized representative surveys – provide different but complementary insights into communities.

How to Interpret the Index and Preliminary Findings

The Fragility Index uses survey questions administered at the household level to measure perceptions of the six dimensions of fragility. While “fragility” is an abstract concept, each dimension is assessed with five concrete survey questions that capture lived experiences relevant to real-world implications of the underlying concept. For example, “Human Fragility” is measured with five survey questions about the respondent household’s: (1) access to food, (2) access to water, (3) access to medical treatment, (4) type of shelter, and (5) exposure to sexual violence.

Figure 1 presents the main results of the Fragility Index; it contains fragility scores for each dimension along the vertical Y-axis and the nine counties along the horizontal X-axis. Darker shades indicate higher fragility scores on a normalized scale from 0 to 100. See Appendix A for discussion on sampling methodology and Appendix B for information about respondent demographics.

Although normalization facilitates visualization of the results on a common scale, it is important to recognize that different types of fragility capture fundamentally different concepts that are not easily comparable. For

15. International Organization for Migration (IOM) and Stanford Immigration Policy Lab, [“Migrant Integration Measurement Toolkit: Offering a Multi-Dimensional Approach to Measure Migrant Integration.”](#) (Geneva, IOM).

16. International Organization for Migration Displacement Tracking Matrix (IOM DTM), [“Solution and Mobility Index - Ghana Round 1,”](#) (Accra, DTM, 2023). International Organization for Migration Displacement Tracking Matrix (IOM DTM), [“Solutions and Mobility Index in Libya,”](#) (Libya, DTM, 2024).

17. International Organization for Migration Displacement Tracking Matrix (IOM DTM), [“Stability Index - DTM Report,”](#) (Dakar, IOM, 2020).

18. International Organization for Migration Displacement Tracking Matrix (IOM DTM), [“Stability Index - Nigeria December 2022-January 2023,”](#) (Asokoro, DTM, 2023).

19. A previous IOM assessment in Iraq comparing the responses of key informants with those of a random sample found significant discrepancies in the opinions of these two populations, likely driven by the fact that the vast majority of key informants were men between the ages of 35-59. See Trigwell, R., J. Philippo-Holmes, E. Zambrano, J. Bahn, P. Hirani and E. Griesmer, [“Validating Humanitarian Data Through Collective Intelligence: A Pilot Study,”](#) (Geneva, IOM, 2022).

20. International Organization for Migration Displacement Tracking Matrix (IOM DTM), [“Solution and Mobility Index - Ghana Round 1,”](#) (Accra, DTM, 2023).

example, Yei county's security and economic fragility scores are both 53. Despite the similarity in scores, the underlying challenges associated with each type of fragility differ significantly, preventing any conclusion that economic and security fragility levels are equivalent in this context. Similarly, in Lainya county, the data should not be interpreted as indicating that the level of environmental fragility (environmental fragility score: 51) is higher than the level of human fragility (human fragility score: 40). Nonetheless, the Fragility Index effectively compares variations within a single dimension of fragility across different counties. For example, Lainya's level of environmental fragility (51) is lower than Rubkona's (66). The Fragility Index can be used to make other comparisons that can inform all three of the primary approaches to aid targeting, discussed below.

Figure 1: Mean fragility by county

Environmental Fragility	48.00	55.00	66.00	57.00	57.00	49.00	51.00	55.00	60.00
Economic Fragility	53.00	46.00	69.00	60.00	53.00	47.00	57.00	59.00	54.00
Societal Fragility	36.00	33.00	32.00	53.00	39.00	31.00	39.00	29.00	29.00
Security Fragility	53.00	73.00	59.00	64.00	58.00	59.00	63.00	61.00	77.00
Political and Legal Fragility	58.00	51.00	64.00	54.00	47.00	58.00	54.00	47.00	61.00
Human Fragility	36.00	45.00	38.00	51.00	39.00	46.00	40.00	39.00	49.00
	Yei	Wau	Rubkona	Pibor	Morobo	Malakal	Lainya	Kajo-keji	Bor South

Implications for Programming and Targeting

- 1. Area-Based:** Comparing relative levels of specific dimensions of fragility across different counties provides valuable insights for programming and targeting. For example, as Figure 1 indicates, Pibor county has the highest level of societal fragility, and Rubkona county has the highest level of political and legal fragility. Wau has the lowest level of economic fragility, and Yei has the lowest level of security fragility. This data would be helpful in the context of deciding which areas to allocate funds designated for a certain purpose.²¹ Similarly, the results can help to identify areas of lower vulnerability where certain types of programming are more likely to succeed. For example, the Business Development programme and other financing programmes for small businesses are more likely to succeed in areas with lower economic fragility, where beneficiaries are more likely to have the necessary skills, resources, and infrastructure to take advantage of the resources. Malakal, for example, has a relatively low level of economic fragility (its economic fragility score is 47 when the range for the nine counties is 46-69); this makes sense given that it was once South Sudan's second largest city. However, transition and recovery programming should also consider other dimensions of fragility. For example, although the county has faced periodic conflicts between different ethnic groups driven by competition over land or other resources, which have disrupted local markets, Malakal's current societal fragility score is relatively low (its societal fragility score is 31 when the range for the nine counties is 29-53).²² Malakal does have a moderate level of security fragility (its security fragility score is 59 when the range for the nine counties is 53-77). This is likely driven in part by concerns about organized crime and gang violence, both of which present risks for business owners.²³ In other words, the design and targeting of programming should take into account not only the most directly relevant dimension of fragility (such as economic fragility in the context of programming to support small businesses) but also other dimensions of fragility (e.g., security, societal, and environmental) that may interact with economic fragility.
- 2. Status-Based:** The Fragility Index results can be analysed for variation in the same dimension of fragility between different sub-groups (e.g., women, IDPs, refugees, persons with disabilities) to inform status-based criteria for allocating aid and programming. For example, Table 1 presents the results of t-tests, a simple test of statistically significant differences between the means of two groups. IDPs and returnees report higher levels of environmental fragility than respondents who have never been displaced ("host communities").²⁴ However, IDPs are not significantly more fragile than host communities in any other dimension. IDPs are also not more fragile than returnees on any dimension of fragility. These tests provide concrete evidence suggesting that ongoing targeting based on IDP status may not be providing support to the most vulnerable residents of South Sudan. Table 1 also suggests that people living outside of county capitals are more environmentally and economically fragile; this finding points to the urgent need to ensure that programming reaches people living in more rural areas. The Fragility Index could also provide data on variation in the vulnerabilities of individuals within a particular

21. Given the lack of reliable population data in South Sudan, the research team refrains from providing suggested sample sizes at the county level in order to achieve a given confidence interval. The Peace and Conflict Resolution Evidence Platform, in a survey enumerated by Detcro, has done the same. See, for example, Jan Pospisil, Melissa Johnston, Adul Alaak Garang, Nyachangkuoth Rambang Tai, "Bring Enough Cows to Marry: Brideprice, Conflict, and Gender Relations in South Sudan," (Coventry and Edinburgh, The University of Edinburgh and PeaceRep and Coventry University, 2024). However, it should be noted that it is not necessary to sample every boma to establish the level of fragility in a given payam, county, or state. What is necessary to achieve reasonably representative samples is to rely on clustered, stratified random sampling. For primers on such methodologies, see, for example, Groves et al, *Survey Methodology*, (Hoboken, Wiley, 2009). Anna Wilke, "10 Things to Know About Sampling," (Berkeley, Evidence in Governance and Politics). Abdul Latif Jameel Poverty Action Lab (J-Pal), "How to Randomize," (Cambridge MA, J-PAL).

22. Radio Tamazuj, "Malakal traders confirm resumption of business, market stability," (Juba, Radio Tamazuj, 2024).

23. Michael Daniel, "100 Youth Arrested on Gang Suspicion in Malakal," (Juba, Eye Radio, 2024).

24. These analyses can also be replicated at more disaggregated geographic levels.

status-based group, helping improve the precision of targeting within that group. For example, it could help identify which women in a given area are most vulnerable to risks such as gender-based violence and would benefit from additional protection assistance.

3. **Individual Household-Level:** The Fragility Index can be used to conduct needs assessments of different types of vulnerability at the household level. This study was conducted with random sample anonymous surveys that are intended to provide representative estimates of population-level characteristics without collecting sensitive personally identifying information about individual respondents. However, humanitarian and development actors can easily adapt the tool to conduct individualized household-level assessments in order to provide more targeted assistance, as is common with cash, shelter, and other in-kind forms of assistance.

Table 1: Differences in Means Across Subgroups²⁵

	IDPs ²⁶ / Host	Returnees / Host	IDPs / Returnees	County Capital / Not
Human Fragility	43.74 / 42.66	42.38 / 42.66	43.74 / 42.38	43.09 / 42.17 °
Political/Legal Fragility	57.61 / 54.93 °	53.88 / 54.93	57.61 / 53.88	55.32 / 54.24 °
Security Fragility	61.35 / 63.21	62.48 / 63.21	61.35 / 62.48	63.44 / 62.43
Societal Fragility	35.42 / 35.75	35.30 / 35.75	35.42 / 35.30	35.56 / 35.69
Economic Fragility	54.90 / 55.31	55.29 / 55.31	54.90 / 55.29	54.56 / 56.03 *
Environmental Fragility	52.67 / 56.21 **	53.30 / 56.21 ***	52.67 / 53.30	54.81 / 55.90 *

25. The following indicate that the differences between the means of the two groups are statistically significant to the following levels: ***p<0.001; **p<0.01; *p<0.05; °p<0.1.

26. IDPs are defined as having moved from one county to another "for security or survival" and not currently living "in the county in which [they] were born." Returnees moved for the same reason but are currently living in the county in which they were born. Defining IDPs and returnees in this fashion means that those who were displaced from their homes but moved to another location (such as a POC site) within the same county are not included. Members of host communities are defined as anyone who does not identify as either an IDP or returnee.

RECOMMENDATIONS AND NEXT STEPS FOR THE FRAGILITY INDEX

This second pilot of the Fragility Index has provided additional evidence of the tool's potential benefits for evidence-based design, implementation, and evaluation of transitional and recovery programming in South Sudan and other fragile contexts. The IOM-Duke research team is seeking additional funding to support the following next steps:

- Expanding the geographical coverage of the Fragility Index to **additional counties** in South Sudan and adapting the tool for use in **other fragile countries**
- Estimating changes in different dimensions of **fragility over time** through subsequent rounds of data collection in the same counties
- Supporting the **dissemination and uptake of the Fragility Index and data** for use by other humanitarian and development actors through IOM's Data for Solutions website, the Humanitarian Data Exchange, and other public platforms
- Improving triangulation, complementarities, data-sharing, and learning between the Fragility Index and other Data for Solutions tools including the MIT, the SI, and the SMI
- Improving random sampling procedures with population density estimates derived from satellite data in the absence of recent reliable census estimates
- Further validation of the Fragility Index and mitigation of potential biases through additional testing by independent third-party research teams
- Using the data to inform testable theories of change for programmatic interventions aimed at reducing or mitigating one or more dimensions of fragility
- Using subsequent rounds of the Fragility Index as baseline and endline assessments to support rigorous impact evaluations of relevant programmatic interventions.

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APPENDIX

A Research Partnership Between IOM South Sudan and Duke University

Preview Prepared for October 30, 2024

Gabriella Levy, Ph.D. and Mara Revkin, Ph.D.¹

1. This summary was prepared by Dr. Gabriella Levy, Assistant Professor of Political Science at the University of Washington and Dr. Mara Revkin, Associate Professor of Law and Political Science at Duke University. Authors are listed alphabetically and contributed equally.

TABLE OF CONTENTS

Appendix A: Sampling methodology.....3

Appendix B: Respondent demographics.....5

Appendix C: Index questions..... 6

Appendix D: Sampled bomas..... 12



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APPENDIX A

Sampling methodology

Bomas in each of the nine counties were primary sampling units (PSUs) and were drawn from the list of bomas contained in IOM's Displacement Tracking Matrix (DTM) 14 Baseline. The nine counties were chosen because these are locations where IOM is currently implementing programming and where DTM has access and the ability to recruit and train local enumerators.¹ More specifically, IOM's DTM team sampled approximately 400 respondents in each county for a total of 3,685 respondents. In each county, the research team randomly selected two bomas in the county capital and two bomas outside the county capital.² The goal was to speak to 100 respondents in each boma. The team also randomly selected a series of backup rural and urban bomas; in case a boma was inaccessible, enumerators were to proceed to pre-selected backup bomas.³ Where bomas did not have 100 households, enumerators completed as many surveys as feasible in the selected bomas and then completed the rest of the surveys in a backup boma.

Within each boma, enumerators followed a random walk procedure. They first confirmed with a community leader or local authority how many dwellings were in the area and then divided that by 100 in order to calculate the appropriate sampling interval. Beginning at a location which enumerators themselves choose, they would then select dwellings separated by the sampling interval to survey. In general, the first identified adult at each selected dwelling was asked to participate. However, enumerators were instructed to oversample men or women as needed within dwellings to achieve a sample of approximately 50 per cent each in each PSU. Participation was anonymous and voluntary, and respondents were able to skip any question.⁴ Enumerators were provided with the questionnaire in English and translated it into the language that the respondent was most comfortable in.⁵

Enumerator training with county focal points was conducted remotely by IOM's DTM team; the focal points then conducted in-person training with their teams. Each county had 10 enumerators, with all of whom DTM had previously engaged. DTM selected a gender-balanced team of enumerators, and all of them were proficient in both English and the predominant local languages.

Prior to enumeration, in each county, DTM staff consulted with government authorities to solicit their support in coordinating and granting access to areas which were to be surveyed.

1. South Sudan Central Bureau of Statistics, "[5th Sudan Population and Housing Census – 2008: Priority Results](#)," (Juba, South Sudan Central Bureau of Statistics, 2009).

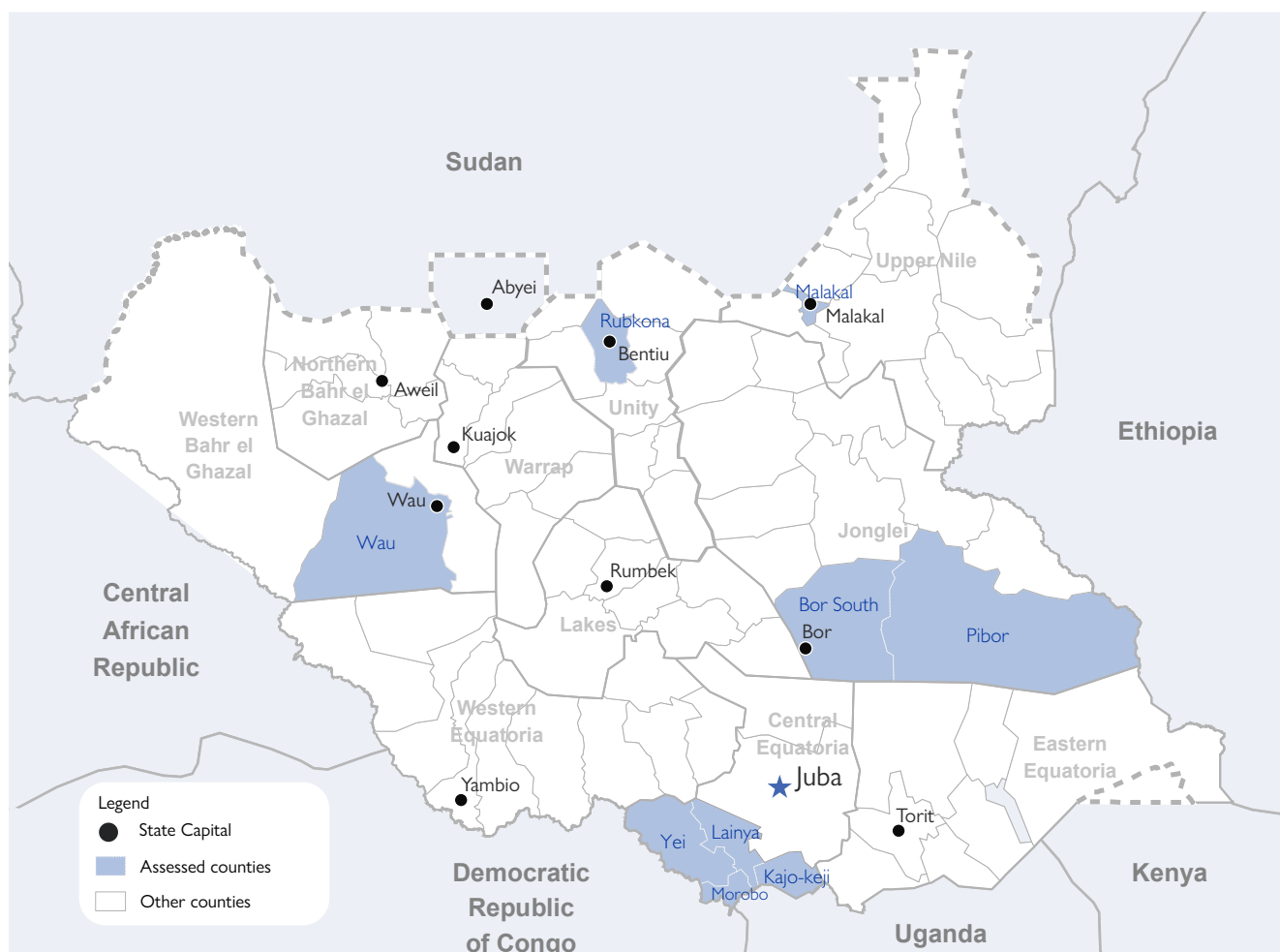
2. See Appendix D for a full list of sampled bomas.

3. There were deviations from this approach, in which replacement bomas were **not** randomly selected, in two out of nine counties. In Pibor, all of the selected bomas were flooded; residents had moved to other, local bomas. The two replacement urban bomas were geographically proximate to each other, but the two replacement rural bomas were geographically proximate to the originally selected bomas. In Rubkona, the two selected rural bomas were not accessible. The replacement rural bomas were geographically proximate to each other.

4. The survey began with a detailed informed consent process in which enumerators explained that declining to participate in the survey would have no impact on their likelihood of receiving assistance from IOM.

5. This approach introduced heterogeneity in the survey instrument presented to respondents. However, translating the survey into every local language before enumeration would be prohibitively expensive.

Map 1: Surveyed counties and payams



The core of the questionnaire consists of 30 questions forming the six fragility indices. Full question text can be found in Appendix C. To construct each of the six indices, each indicator value was normalized to range from 0-1; 0 indicates the lowest possible level of fragility and 1 indicates the highest possible level of fragility.⁶ Then, six fragility scores were calculated for each subject; each fragility score is a simple additive index of the five questions which make up that that score.⁷ In other words, each question is weighted equally in the indices. As a final step, the indices were rescaled to range from 0-100; 0 indicates the lowest observed fragility score and 100 indicates the highest observed fragility score.⁸

6. More formally, the normalization formula is as follows: $(\text{value} - \text{minimum possible value}) / (\text{maximum value} - \text{minimum possible value})$. Note that all responses with negative values (e.g. “-2. Prefer not to answer”) were changed to NA before this normalization. To ensure that higher values indicate higher levels of fragility, many variables were reverse coded; for details, see Appendix C.

7. To avoid scores that were artificially lowered due to missing values, for all questions which contributed to indices, the research team imputed missing values with the median response values to those questions before building the additive indices. As a result, fragility scores contain imputed data. However, analysis of all individual survey questions uses unimputed versions of the variables.

8. More formally, this normalization and rescaling formula is as follows: $[(\text{score} - \text{minimum actual score}) / (\text{maximum actual score} - \text{minimum actual score})] \times 100$.

APPENDIX B

Respondent demographics

State	County	Number of Respondents	Gender Distribution	% IDPs or Refugees	% Returnees	3 Largest Ethnicities (% of Respondents) ⁹
Central Equatoria	Kajo-Keji	398	53% Female 47% Male	1%	44%	Kuku (60%), Bari (38%), Bviri (1%)
Central Equatoria	Yei	410	50% Female 50% Male	3%	16%	Kakwa (57%), Baka (11%), Other (5%),
Central Equatoria	Lainya	414	41% Female 59% Male	8%	27%	Bari (98%), Kakwa (1%), Bviri (0%)
Central Equatoria	Morobo	404	51% Female 49% Male	4%	54%	Kakwa (55%), Bari (23%), Other (18%)
Jonglei	Bor South	429	71% Female 29% Male	5%	29%	Dinka (100%), Baka (0%), Mundari (0%) ¹⁰
Jonglei	Pibor	407	66% Female 34% Male	4%	12%	Murle (93%), Anyuak (3%), Dinka (1%)
Western Bahr el Ghazal	Wau	402	64% Female 36% Male	2%	11%	Fertit (52%), Other (30%), Bviri (5%)
Upper Nile	Malakal	421	73% Female 27% Male	11%	22%	Shilluk (52%), Dinka (42%), Nuer (5%)
Unity	Rubkona	400	84% Female 16% Male	1%	4%	Nuer (97%), Other (2%) ¹¹
Total (all 9 counties)		Total: 3,685 ¹²	61% Female 39% Male	4%	24%	Bari (19%), Dinka (18%), Kakwa (13%)

The sample includes a small population of refugees who indicate that they were displaced from locations outside of South Sudan. 16 are from Sudan, 13 are from Uganda, 4 are from the Democratic Republic of the Congo, 1 is from the Central African Republic, 1 is from Ethiopia, and 1 is from Kenya.

9. Respondents were provided with the following options: Acholi, Anyuak, Azande, Baka, Bari, Bongo, Bviri, Didinga, Dinka, Dungotona, Fertit, Kakwa, Kuku, Lango, Lndi, Mundari, Murle, Ndogo, Nuer, Shilluk, and Other.

10. There was 1 Baka respondent and 1 Mundari respondent in Bor South.

11. The Murle, Shilluk, and Lndi groups each represent just .25% of respondents in the county.

12. There are 13 respondents for whom the location was not recorded. These are not included in the total number of respondents and have been removed from the data and analysis.

APPENDIX C

Index questions

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Human Fragility [Health]	Food Access	Q5.5 Over the past year, how often, if ever, have you or anyone in your family gone without the following items: never, just once or twice, several times, many times, or always? → Q5.5.1 Enough food to eat 0. Never 1. Just once or twice 2. Several times 3. Many times 4. Always -2. Prefer not to answer	NA	No
	Water Access	Q5.5 Over the past year, how often, if ever, have you or anyone in your family gone without the following items: never, just once or twice, several times, many times, or always? → Q5.5.2 Enough clean water for home use 0. Never 1. Just once or twice 2. Several times 3. Many times 4. Always -2. Prefer not to answer	NA	No
	Medical Treatment Access	Q5.5 Over the past year, how often, if ever, have you or anyone in your family gone without the following items: never, just once or twice, several times, many times, or always? → Q5.5.3 Necessary medicines or medical treatment 0. Never 1. Just once or twice 2. Several times 3. Many times 4. Always -2. Prefer not to answer	NA	No
	Shelter	Q7.1 What type of shelter are you currently living in? 1. No shelter 2. Rabooka 3. Tukul 4. Bricked structure 5. Other, specify^[1] -2. Prefer not to answer	Replacement of shelter types with culturally appropriate ones	Yes
	Sexual violence	Q8.5 On a scale from 1 to 5, where 1 means “very unfrightened” and 5 means “very frightened,” how afraid are you of the following happening in your community? → Q8.5.6 Sexual violence 1. Very unfrightened 2. Unfrightened 3. Neither frightened nor unfrightened 4. Frightened 5. Very frightened -2. Prefer not to respond	NA	No

^[1] This answer was converted into NA

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Economic Fragility	Education	Q2.3 What is the highest level of schooling you have completed? - No formal schooling - Primary school - Secondary school - Vocational courses only - University - Prefer not to answer	NA	Yes
	Financial satisfaction	Q5.6 From 1 to 5, where 1 is “very dissatisfied” and 5 is “very satisfied,” how satisfied are you with your household’s financial situation? - Very dissatisfied - Dissatisfied - Neither dissatisfied nor satisfied - Satisfied - Very satisfied - Prefer not to answer	NA	Yes
	Ability to pay for expected large costs	Q5.7 On a scale from 1 to 5, where 1 is “very difficult” and 5 is “very easy,” how easy would it be for your family to pay a bride price currently? - Very difficult - Difficult - Neither difficult nor easy - Easy - Very easy - Prefer not to answer	Replacement of “bride price” with locally relevant expected large cost	Yes
	Ability to pay for unexpected large costs	Q5.9 How severe would you say the impact of an unexpected death would be on your household’s economic situation? - Not severe at all (no impact) - Not severe - Somewhat severe - Severe - Extremely severe - Prefer not to answer	Replacement of “unexpected death” with locally relevant unexpected large cost	No
	Dependency on aid	Q5.22 To what extent is your household dependent on monetary or other forms of aid from NGOs, the UN, or the government? - No dependency - Low dependency - Moderate dependency - High dependency - Full dependency - Prefer not to answer	NA	No

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Environmental Fragility	Climate change's impact on income	Q11.10A Many people in South Sudan have experienced the following: too much rain, too little rain, temperatures that are too high, temperatures that are too low, pests and crop diseases, and deforestation and erosion. How severe would you say the impact of any of these experiences have been on your household income, on a scale from 1 to 5 where 1 means “not severe at all and 5 means “extremely severe?” 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	Replacement of “South Sudan” with relevant country and list of climate hazards with locally relevant ones	No
	Climate change's impact on food consumption	Q11.10B And how severe would you say the impact of any of these experiences have been on your household food consumption? 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	NA	No
	Climate change's impact on mobility	Q11.10C And how severe would you say the impact of any of these experiences have been on your household's ability to choose where you live? 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	NA	No
	Climate change's impact on health	Q11.10D And how severe would you say the impact of any of these experiences have been on your household's health? 1. Not severe at all (no impact) 2. Not severe 3. Somewhat severe 4. Severe 5. Extremely severe -2. Prefer not to answer	NA	No
	Ability to act on early warning information to ensure household safety	- Q11.11 Imagine that there was a flood coming to your town tomorrow. If you knew about this incoming flood before the water arrived, to what extent do you agree with the following statement: “I would be able to use this information to protect my household?” 1. Strongly disagree 2. Disagree 3. Neither agree nor disagree 4. Agree	Replace “flood” with locally plausible climate change-induced emergency	Yes

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Societal Fragility	Experiences of acceptance	Q6.1 How much do you feel welcomed or accepted in your current community? - Not at all accepted - A little accepted - Accepted - Very accepted -2. Prefer not to answer	NA	Yes
	Experiences of discrimination	Q6.2 In the past year, how often, if at all, have you been personally discriminated against based on any of the following: your gender, your religion, your ethnicity, any disability you might have, or your place of origin? - Never - Sometimes - Often - Always -2. Prefer not to answer	NA	No
	Social cohesion	Q6.3 To what extent do you agree with the following statement? "People in your boma are willing to help each other." - Strongly disagree - Disagree - Neither agree nor disagree - Agree - Strongly agree -2. Prefer not to answer	Replace "boma" with locally relevant geographic boundary	Yes
	Network embeddedness	Q6.4 Imagine that you were in an argument with your neighbor about a small issue. To what extent do you agree or disagree with the following statement? "I know who I could seek help from to resolve this argument." - Strongly disagree - Disagree - Neither agree nor disagree - Agree - Strongly agree -2. Prefer not to answer	NA	Yes
	Migrant-Local Cohesion	Q6.9 How would you describe the relationship between individuals who are not originally from the community and the rest of the community members? - Very poor - Poor - Neutral - Good - Very good -2. Prefer not to answer -3. Not applicable^[2]	NA	Yes

Dimension	Concept	Questions in S Sudan	Modifications for Other Contexts	Reverse Coded
Legal/Political Fragility	Trust in state government	Q10.1 How much confidence do you have in the following institutions? → Q10.1.2 Your state government - No confidence - Some confidence - Confidence - A lot of confidence -2. Prefer not to answer	NA	Yes
	Trust in informal authorities	Q10.1 How much confidence do you have in the following institutions? → Q10.1.10 Informal authorities, such as councils of elders - No confidence - Some confidence - Confidence - A lot of confidence -2. Prefer not to answer	Replace “councils of elders” with locally relevant informal authorities	Yes
	Participation in democracy without fear	Q10.10 South Sudan’s next national elections are scheduled for December 2024 after being postponed several times. How safe or unsafe would you feel going to a polling station to cast your vote? - Very unsafe - Somewhat unsafe - Neither safe nor unsafe - Somewhat safe - Very safe -2. Prefer not to answer	Replace first sentence with relevant details about upcoming national election	Yes
	Freedom of speech	Q10.12 At present, how safe do you feel speaking publicly on politically sensitive issues? - Very unsafe - Unsafe - Neither safe nor unsafe - Safe - Very safe -2. Prefer not to answer	NA	Yes
	Trust in national government	Q10.13 Which of the following statements best describes the role that politicians in Juba play in your community? - Politicians in Juba make this community much less safe - Politicians in Juba make this community less safe - Politicians in Juba neither make this community more nor less safe - Politicians in Juba make this community more safe - Politicians in Juba make this community much more safe -2. Prefer not to answer	Replace “Juba” with national capital	Yes

APPENDIX D

Sampled Bomas

County	Payam	Boma	Capital	In Original Sampling Plan	# of Respondents
Bor South	Baidit	Mathiang	0	Yes	105
Bor South	Jalle	Akuai-deng	0	Yes	103
Bor South	Bor	Panapet	1	Yes	113
Bor South	Bor	Langbar	1	Yes, backup	108
Kajo-keji	Nyepo	Kansuk	0	Yes	91
Kajo-keji	Nyepo	Lori	0	Yes, backup	102
Kajo-keji	Lire	Mekir	1	Yes	97
Kajo-keji	Lire	Likamerok	1	Yes	87
Kajo-keji	Lire	Mere	1	Yes, backup	21
Lainya	Mukaya	Dimu One	0	Yes	77
Lainya	Kopera	Korobe	0	Yes	100
Lainya	Mukaya	Biri	0	Yes, backup	27
Lainya	Lainya	Lokurubang	1	Yes	122
Lainya	Lainya	Bereka	1	Yes	88
Malakal	Lelo	Obwa	0	Yes	51
Malakal	Ogod	Wau	0	Yes	77
Malakal	Ogod	Padit	0	Yes, backup	73
Malakal	Malakal North	Hai Chathi	1	Yes	124
Malakal	Malakal South	Tarawa	1	Yes	26
Malakal	Malakal Centre	Mederia	1	Yes, backup	70
Morobo	Gulumbi	Kendila	0	Yes	104
Morobo	Lujulo	Yugufe	0	Yes	100
Morobo	Kimba	Kaya	1	Yes	100
Morobo	Kimba	Kimba Centre	1	Yes	100
Pibor	Lekuangle	Lekuangle	0	No	102
Pibor	Gumruk	Tangnyang	0	No	101
Pibor	Pibor	Kavachoch	1	No	101
Pibor	Pibor	Pibor	1	No	103
Rubkona	Nhialdiu	Nyaromne	0	No	121
Rubkona	Panhiany	Panhiany	0	No	80
Rubkona	Rubkona	Nying	1	Yes	29

County	Payam	Boma	Capital	In Original Sampling Plan	# of Respondents
Rubkona	Rubkona	Tong	1	Yes	50
Rubkona	Rubkona	Yoanyang	1	Yes, backup	120
Wau	Bagari	Bringi	0	Yes	50
Wau	Bagari	Bussere	0	Yes	152
Wau	Wau North	Hai Kalvario	1	Yes	100
Wau	Wau South	Nazareth	1	Yes	100
Yei	Mugwo	Longamere	0	Yes	94
Yei	Tore	Adio	0	Yes	100
Yei	Yei Town	Hai Sobe Zone	1	Yes	116
Yei	Yei Town	Pakkula	1	Yes	100



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